

APFD

Signal-Based Ranking on Existing Tools

Top-20% Bugs (%)

COMET	0.348	0.403	0.553	0.345	0.508	0.375	0.548
DevMuT	0.676	0.470	0.561	0.555	0.472	0.573	0.561
ModelMeta	0.406	0.567	0.653	0.409	0.654	0.559	0.653
Muffin	0.162	0.498	0.470	0.473	0.494	0.494	0.470
NEURI	0.488	0.491	0.488	0.521	0.437	0.495	0.529
GenCoG	0.461	0.500	0.461	0.474	0.525	0.461	0.561
LEMON	0.493	0.499	0.549	0.545	0.545	0.545	0.545
Gandalf	0.502	0.481	0.482	0.482	0.482	0.482	0.487
NNSmith	0.343	0.500	0.533	0.403	0.540	0.527	0.551
	Original	Random	Node Count	Edge Count	Depth	Operator Count	OTC



COMET	18.2	17.9	24.3	6.2	20.8	8.8	24.1
DevMuT	14.6	17.9	30.8	26.9	15.4	28.8	30.8
ModelMeta	15.7	22.2	35.8	0.6	41.8	12.7	36.3
Muffin	12.3	22.0	18.0	17.8	15.2	20.7	18.0
NEURI	19.8	18.6	19.8	26.1	15.4	20.7	22.2
GenCoG	17.2	21.0	17.2	16.7	22.9	17.2	25.0
LEMON	23.9	16.9	24.3	24.3	24.3	24.3	24.2
Gandalf	21.2	17.5	17.5	17.5	17.5	17.5	17.6
NNSmith	14.9	18.8	14.9	2.5	13.4	14.9	16.9
	Original	Random	Node Count	Edge Count	Depth	Operator Count	OTC



First Bug Rank (%)

COMET	0.1	0.1	3.0	6.5	0.1	0.2	0.0
DevMuT	2.0	1.0	2.0	2.0	3.0	2.0	1.0
ModelMeta	6.6	0.0	14.5	14.5	0.0	12.9	0.0
Muffin	0.0	0.0	0.2	0.1	0.1	0.3	0.2
NEURI	18.1	21.1	13.7	16.3	23.6	18.1	14.5
GenCoG	13.3	13.2	17.1	16.1	10.3	13.3	10.5
LEMON	33.7	25.5	27.8	27.8	27.8	27.8	27.7
Gandalf	4.6	5.5	7.0	7.0	7.0	7.0	6.7
NNSmith	45.4	53.1	51.2	60.3	45.3	45.4	42.9
	Original	Random	Node Count	Edge Count	Depth	Operator Count	OTC



Spearman Rho vs. Bug Label

COMET	0.011	0.012	-0.266	-0.313	0.092	-0.255	0.117
DevMuT	-0.132	0.005	0.151	0.151	-0.192	0.261	0.219
ModelMeta	0.204	-0.011	-0.233	-0.233	0.457	0.156	0.412
Muffin	-0.020	-0.044	-0.076	-0.069	-0.024	-0.015	-0.076
NEURI	-0.001	0.008	0.056	0.018	-0.039	0.000	0.019
GenCoG	0.080	-0.005	-0.034	-0.051	0.062	0.000	0.132
LEMON	0.042	-0.056	0.089	0.089	0.089	0.089	0.102
Gandalf	-0.008	-0.019	-0.070	-0.070	-0.070	-0.070	-0.052
NNSmith	-0.012	0.013	-0.023	-0.051	0.016	0.000	0.023
	Original	Random	Node Count	Edge Count	Depth	Operator Count	OTC

